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BLUETOOTH-CONTROLLED ROBOT
Using The STM32 Blue Pill

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Building a robot is an effective way to understand embedded systems, control logic, and real-world electronics integration. This system demonstrates a Bluetooth-controlled mobile robot built around the low-cost STM32 Blue Pill microcontroller. It features an OLED display, four BO motors, and Li-ion batteries for power.

The robot can be controlled wirelessly from a smartphone via Bluetooth, while the OLED displays real-time system information such as connection status, eye animations, and battery level.

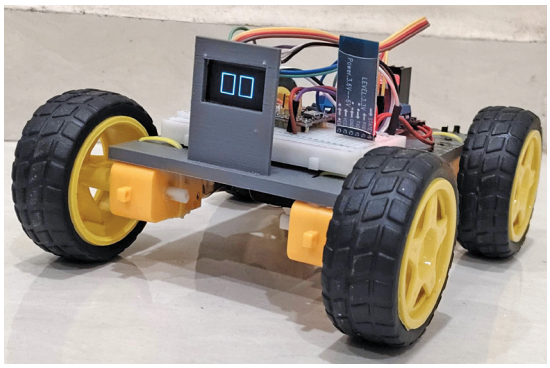


Fig. 1: Authors' prototype

With its low-cost design, compact size, and simple connections, the robot can serve as an educational and experimental platform for students and hobbyists. It demonstrates the practical implementation of key concepts such as GPIO interfacing, serial communication, motor-driver control, display interfacing, and battery-powered operation.

The system can be further enhanced by integrating sensors, obstacle detection, and other robotic applications. Fig. 1 shows the authors' prototype, while Fig.

BILL OF MATERIALS

Name	Specification	Quantity
STM32	Blue Pill board	1
Battery	Li-ion, 3.7V, 2000mAh-2500mAh	2
Battery holder	Two-cell battery holder	1
L298N	Motor driver	1
HC-05	Bluetooth module	1
BO gear motor	Dual shaft 200rpm	4
OLED display	0.96-inch (2.4cm) I2C	1
Switch	SPST	1

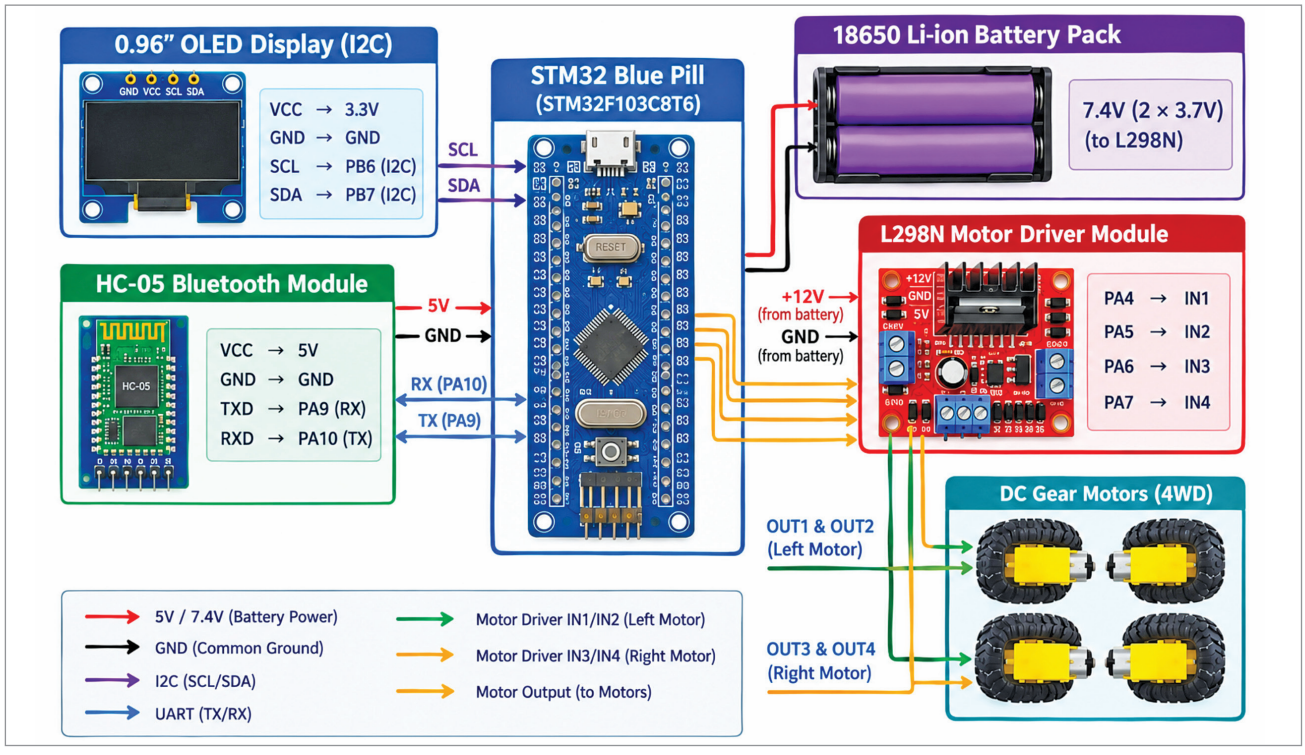


Fig. 2: Circuit connections with STM32 Blue Pill